P.Vishwa Nandan
192473015

Experiment 7: Customer Order Analysis Using Pandas

Code:

```
import pandas as pd


order_data = pd.DataFrame({
    'Customer_ID': [101, 102, 101, 103],
    'Order_Date': ['2026-01-10', '2026-01-12', '2026-01-15', '2026-01-20'],
    'Product': ['Pen', 'Book', 'Pen', 'Book'],
    'Quantity': [2, 4, 6, 8]
})


print(order_data.groupby('Customer_ID').size())
print(order_data.groupby('Product')['Quantity'].mean())


order_data['Order_Date'] = pd.to_datetime(order_data['Order_Date'])
print("Earliest:", order_data['Order_Date'].min())
print("Latest:", order_data['Order_Date'].max())
```

Output:

```
Customer_ID
101    2
102    1
103    1
dtype: int64
Product
Book    6.0
Pen     4.0
Name: Quantity, dtype: float64
Earliest: 2026-01-10 00:00:00
Latest: 2026-01-20 00:00:00
```

Experiment 8: Top 5 Best-Selling Products Using Pandas

Code:

```
import pandas as pd


sales_data = pd.DataFrame({
    'Order_Date': ['2026-10-01', '2026-10-02', '2026-10-03',
                   '2026-10-04', '2026-10-05', '2026-10-06'],
    'Product': ['Pen', 'Book', 'Pen', 'Bag', 'Book', 'Pencil'],
    'Quantity': [10, 8, 15, 5, 12, 7]
})


sales_data['Order_Date'] = pd.to_datetime(sales_data['Order_Date'])


top_products = sales_data.groupby('Product')['Quantity'].sum()
print(top_products.sort_values(ascending=False).head(5))
```

Output:

```
Product
Pen       25
Book      20
Pencil     7
Bag        5
Name: Quantity, dtype: int64
```

Experiment 9: Real Estate Property Analysis Using Pandas

Code:

```
import pandas as pd


property_data = pd.DataFrame({
    'Property_ID': [1, 2, 3, 4],
    'Location': ['Chennai', 'Madurai', 'Chennai', 'Salem'],
    'Bedrooms': [3, 5, 4, 6],
    'Area': [1200, 2000, 1500, 2500],
    'Price': [5000000, 7000000, 6000000, 8000000]
})


print(property_data.groupby('Location')['Price'].mean())
print("More than 4 bedrooms:",
      len(property_data[property_data['Bedrooms'] > 4]))


print("Largest property:")
print(property_data.loc[property_data['Area'].idxmax()])
```

Output:

```
Location
Chennai    5500000.0
Madurai    7000000.0
Salem      8000000.0
Name: Price, dtype: float64
More than 4 bedrooms: 2
Largest property:
Property_ID          4
Location         Salem
Bedrooms             6
Area              2500
Price          8000000
Name: 3, dtype: object
```